Microsoft

Theoria cum Praxi

AZ-900

Learning path 2: Azure architecture and services

Agenda

AZ-900

Learning path 1: Cloud concepts

Learning path 2: Azure architecture and services

Learning path 3: Azure management and governance

Describe the core architectural components of Azure

Kaufhaus

What is Microsoft Azure?

New Portal: Foundry

Microsoft Azure Service Categories

Compute
- Virtual Machines
- App Service
- Azure Functions
- Docker Container Apps
- Azure Kubernetes Service

Networking
- Virtual Network
- Load Balancer
- VPN Gateway
- ExpressRoute
- Azure DNS

Storage
- Blob Storage
- Azure Files
- Queue Storage
- Table Storage
- Managed Disks

Databases
- Azure SQL
- Azure Cosmos DB
- PostgreSQL Flex
- MySQL Flex
- Redis Cache

AI + ML
- Azure OpenAI
- AI Services
- Machine Learning
- Bot Service
- AI Search

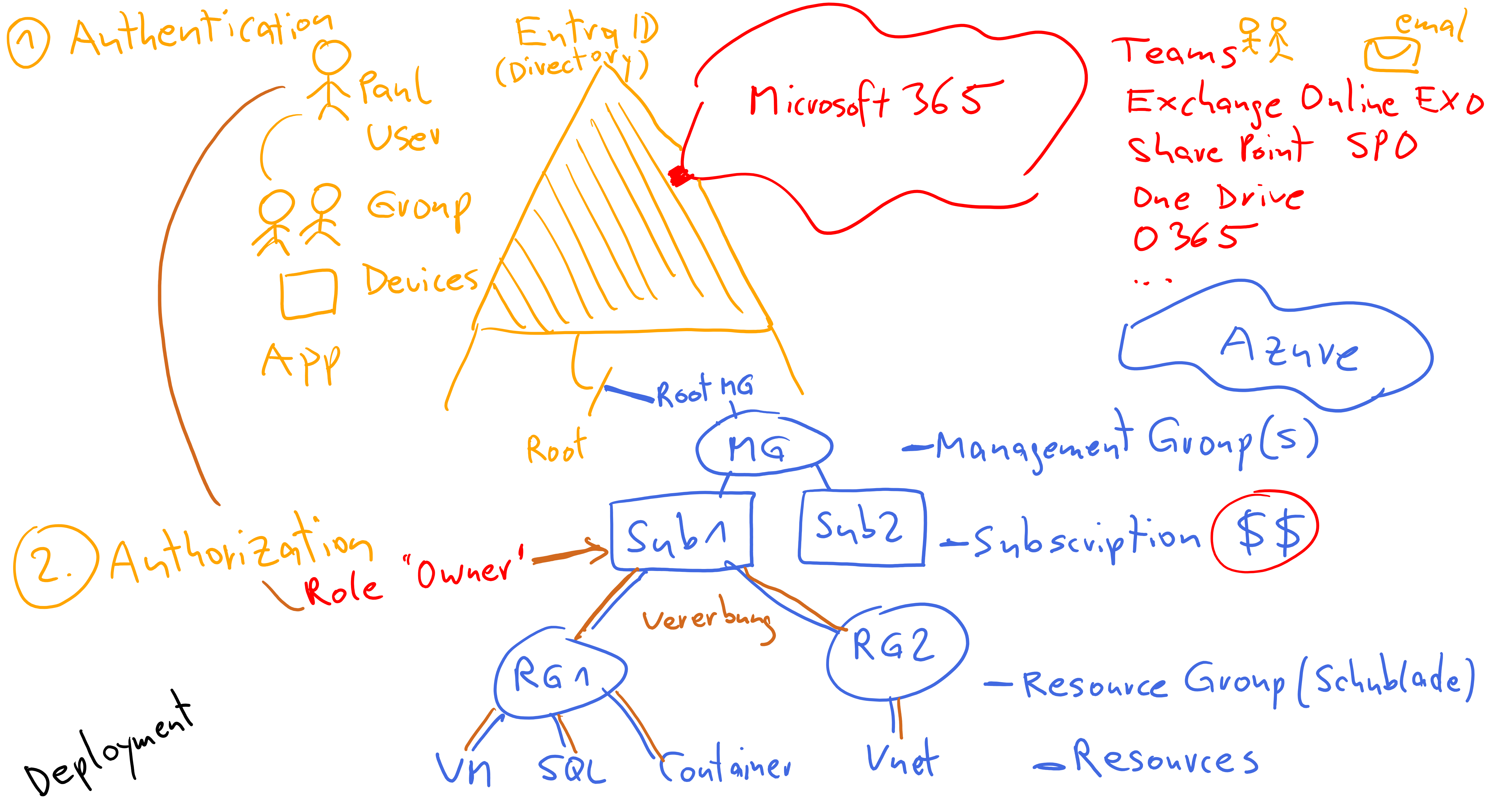
Identity + Security
- Microsoft Entra ID
- Key Vault
- Defender for Cloud
- DDoS Protection
- Firewall

DevOps + Management
- Azure DevOps
- Azure Monitor
- ARM Templates
- Azure Policy
- Cost Management

IoT
- IoT Hub
- IoT Central
- Digital Twins
- Stream Analytics
- Event Hubs

Analytics
- Synapse Analytics
- Data Factory
- Databricks
- HDInsight
- Azure Data Explorer

Integration
- Logic Apps
- API Management
- Service Bus
- Event Grid
- Notification Hubs



1 Authentication
Paul
User
Group
Devices
App
Entra ID
(Directory)
Root
Root MG
Microsoft 365
Teams
email
Exchange Online EXO
Share Point SPO
One Drive
O365
...
Azure
MG
Management Group(s)
Sub1
Sub2
Subscription
$$
2. Authorization
Role "Owner"
Vererbung
RG1
RG2
Resource Group (Schublade)
VM
SQL
Container
Vnet
Resources
Deployment

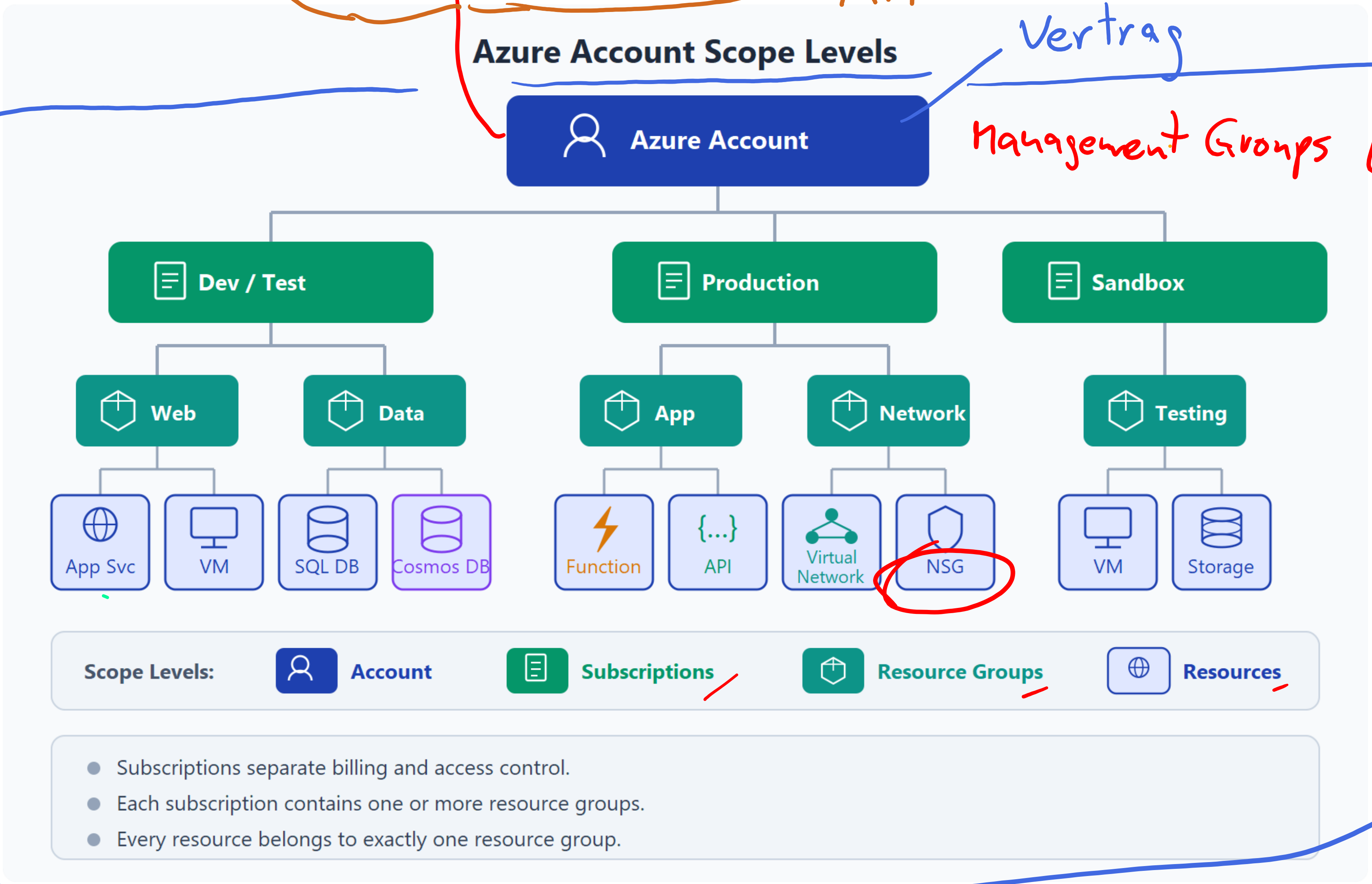
Get started with Azure accounts

Entra ID
M365

Tenant
API Microsoft Graph

Azure Resource Manager
ARM
API

Vertrag



Free

Management Groups (6fach)

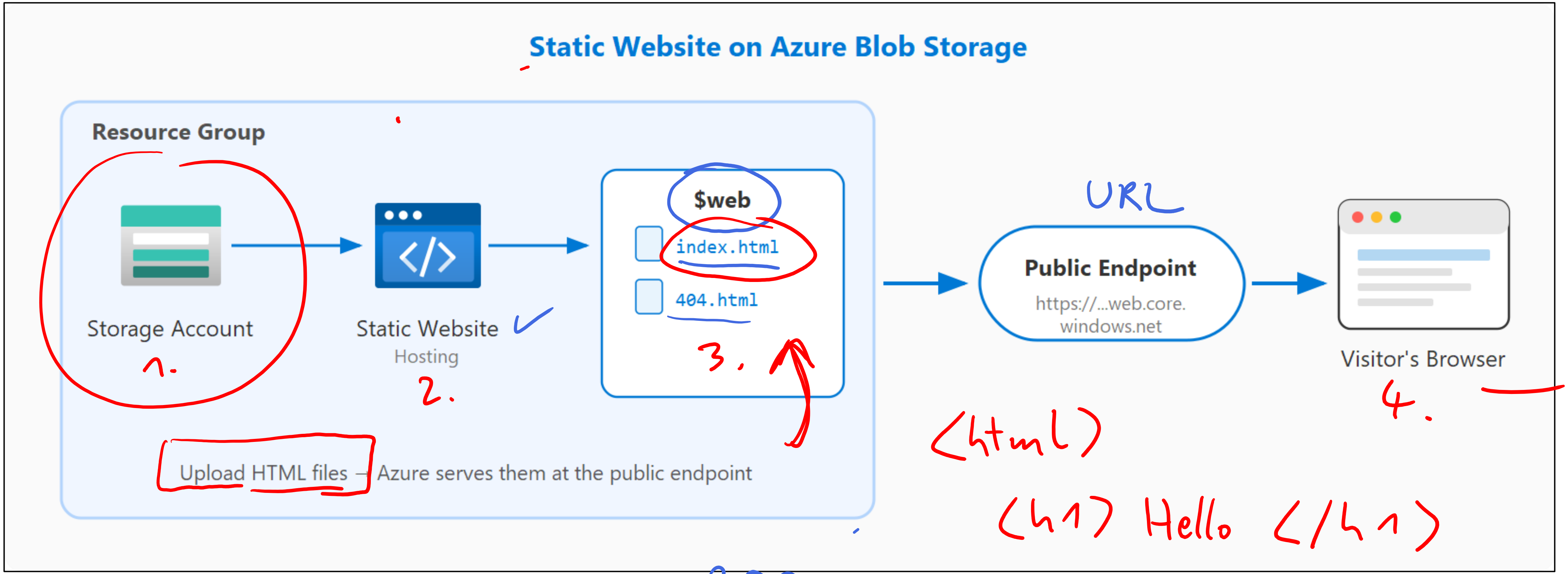
Subscription

Resource Group (flat)

Resources

AZ-900 Lab on Skillable

Project: Static website on Blob Storage diagram



<html>

<h1> Hello </h1>

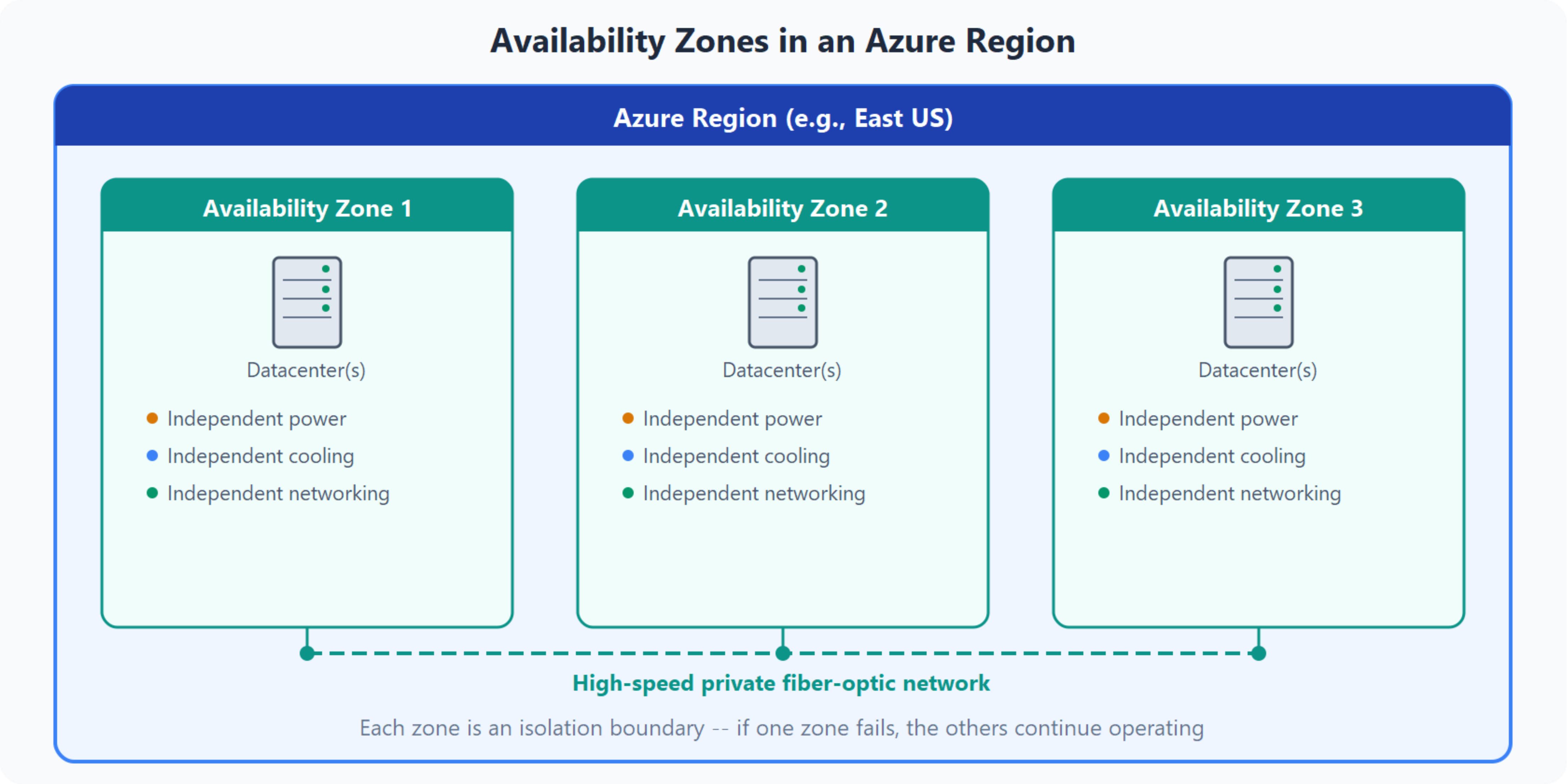
www 1985

Tim Berners-Lee CERN

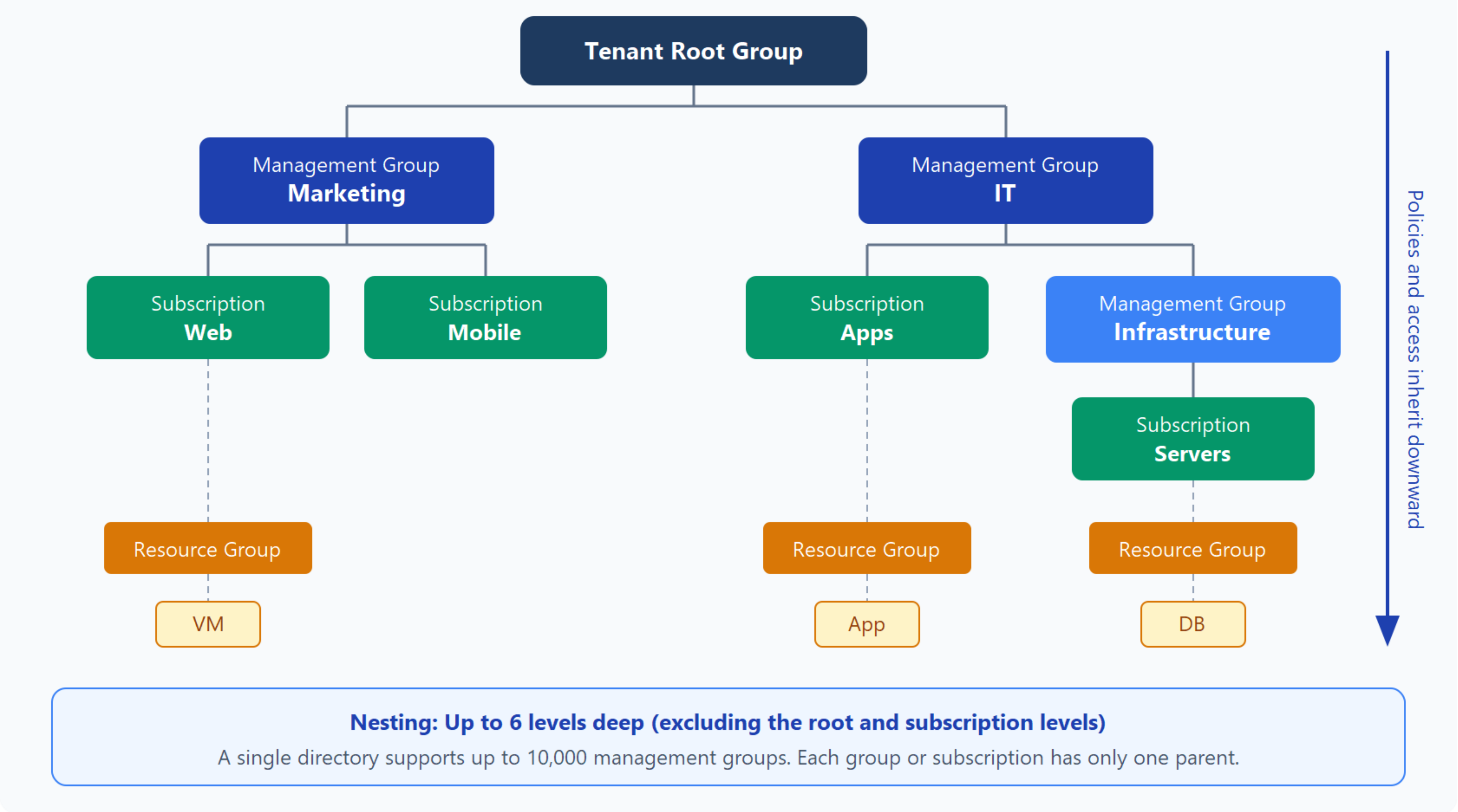
200
404
500

</html>

Azure physical infrastructure



Azure management infrastructure



Knowledge check

1 **Which statement is true about Azure regions and availability zones?**

☐ Every region has exactly 3 zones

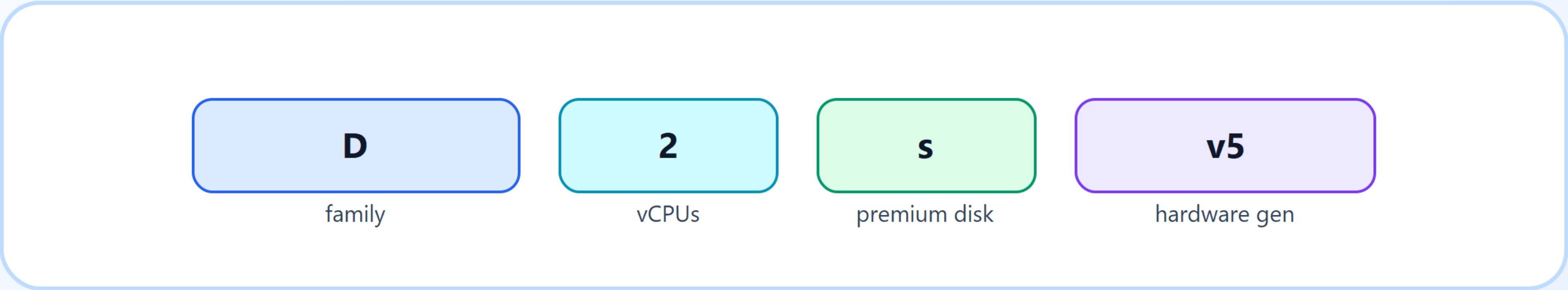
☐ Availability zones are separate datacenters within a region

☐ Regions are only logical groupings, not physical

Describe Azure compute services

Decoding Azure Virtual Machine names

How to read Standard_D2s_v5



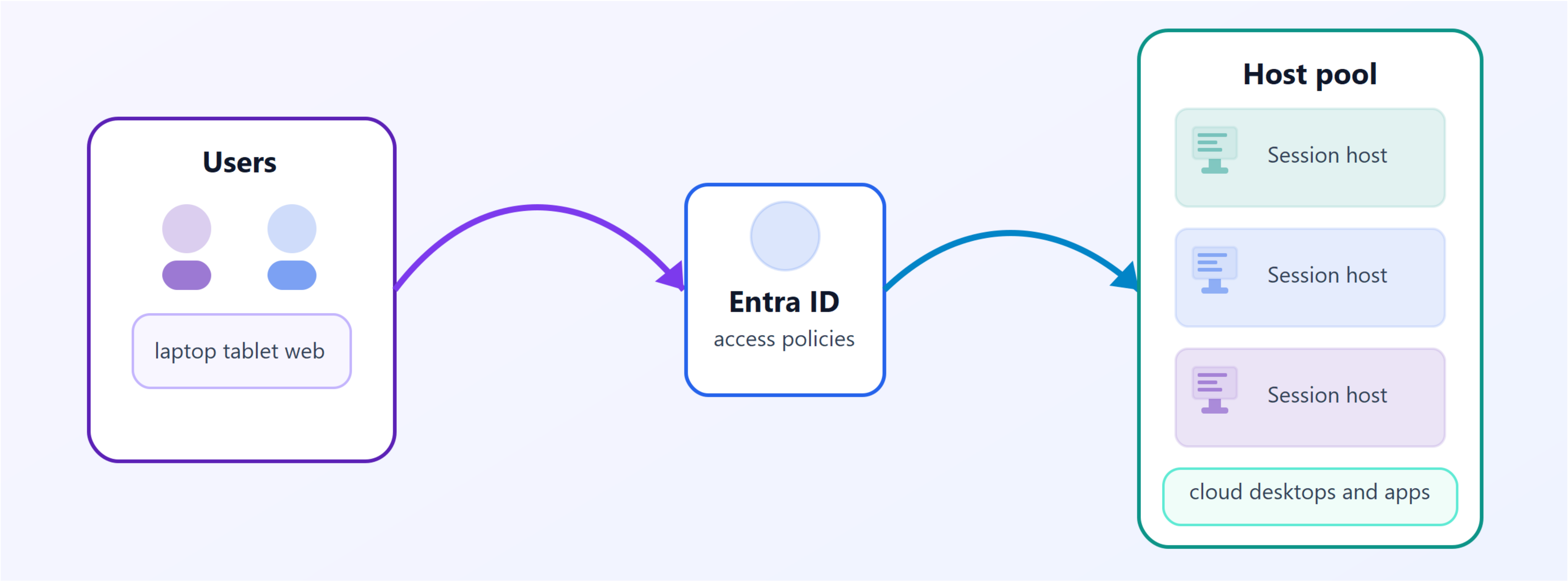
Azure Virtual Machines families

VM size families

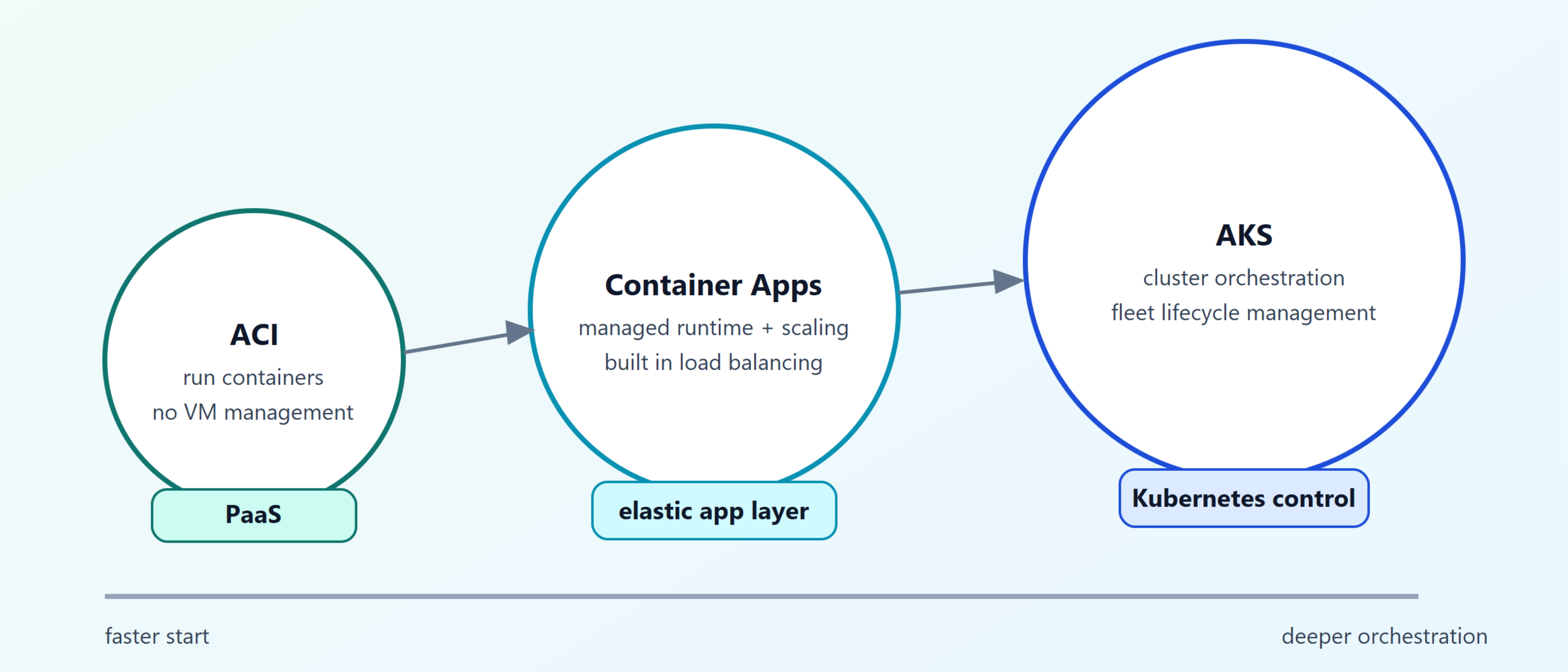
Choose the family that best matches workload behavior.

- **B** burstable
- **D** general
- **E** memory
- **F** compute
- **M** large memory
- **L** storage
- **N** GPU

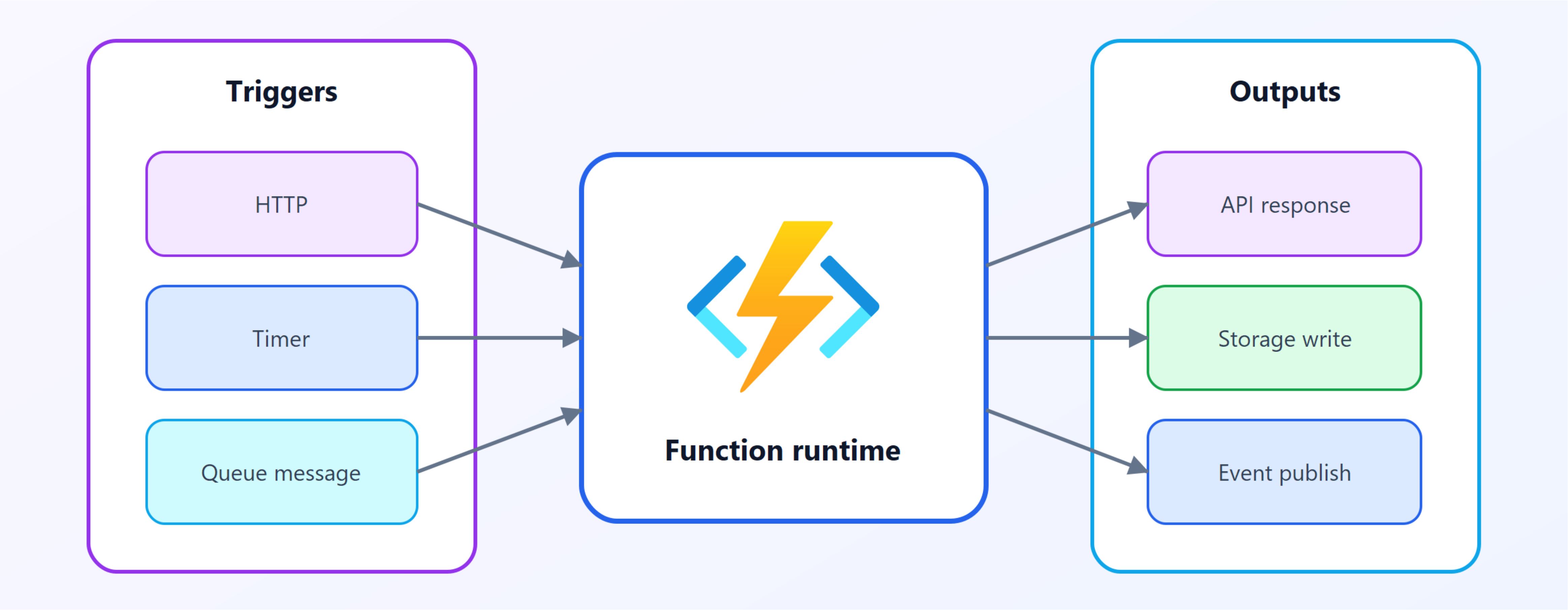
Azure Virtual Desktop



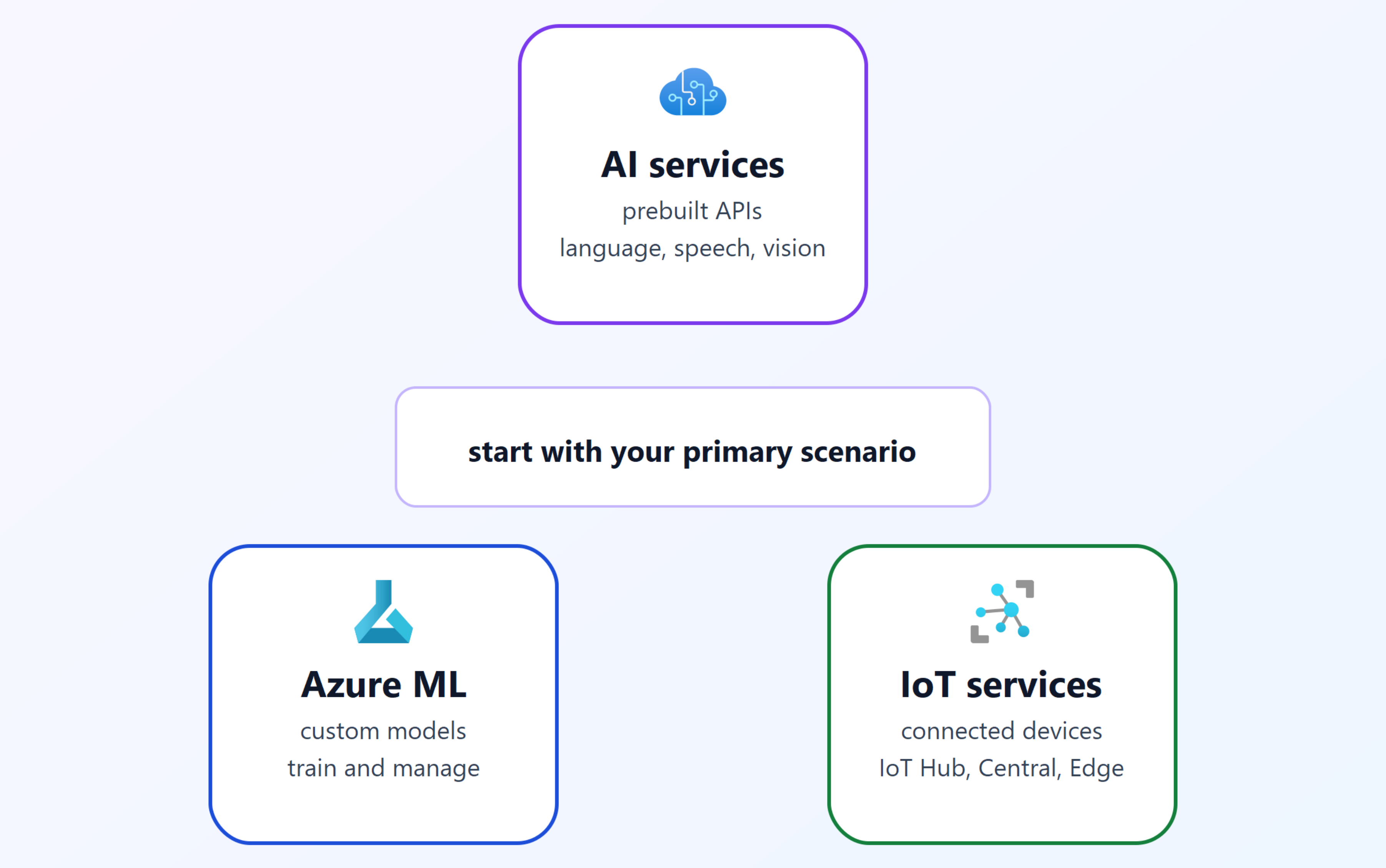
Azure container services



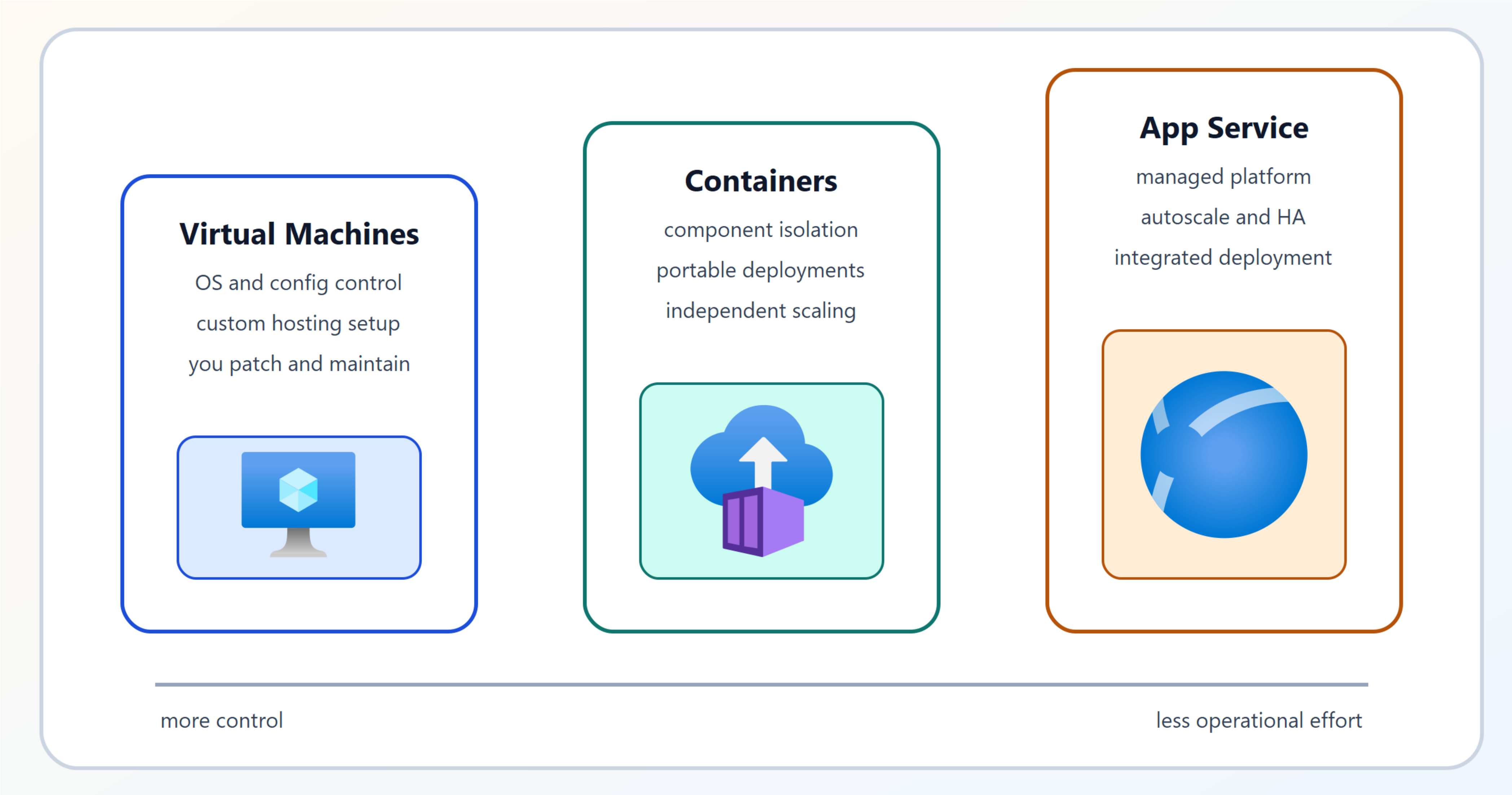
Azure Functions



AI, machine learning, and IoT and Edge services



Application hosting options



Knowledge check

1 **Which Azure service is best when you want to run containerized applications without managing VM infrastructure?**

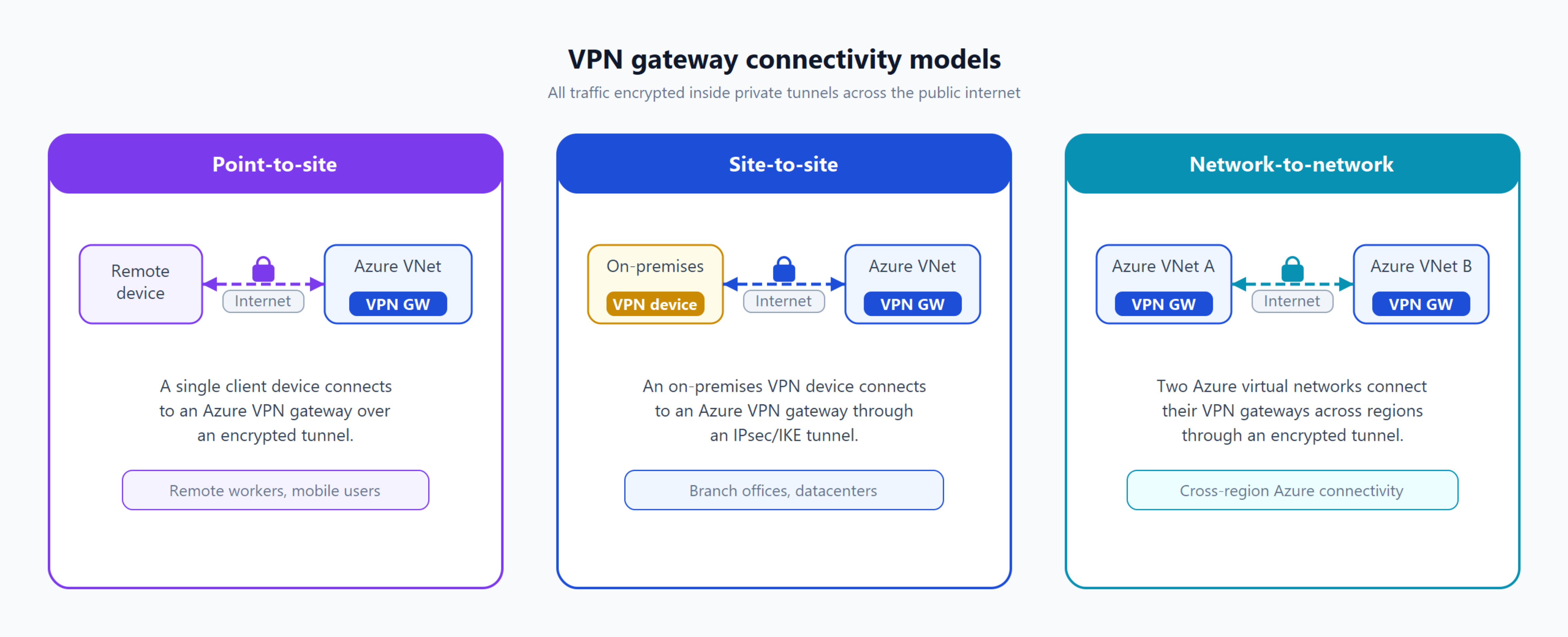
- ☐ Azure Virtual Machines
- ☐ Azure Functions
- ☐ Azure Kubernetes Service (AKS)

Describe Azure networking services

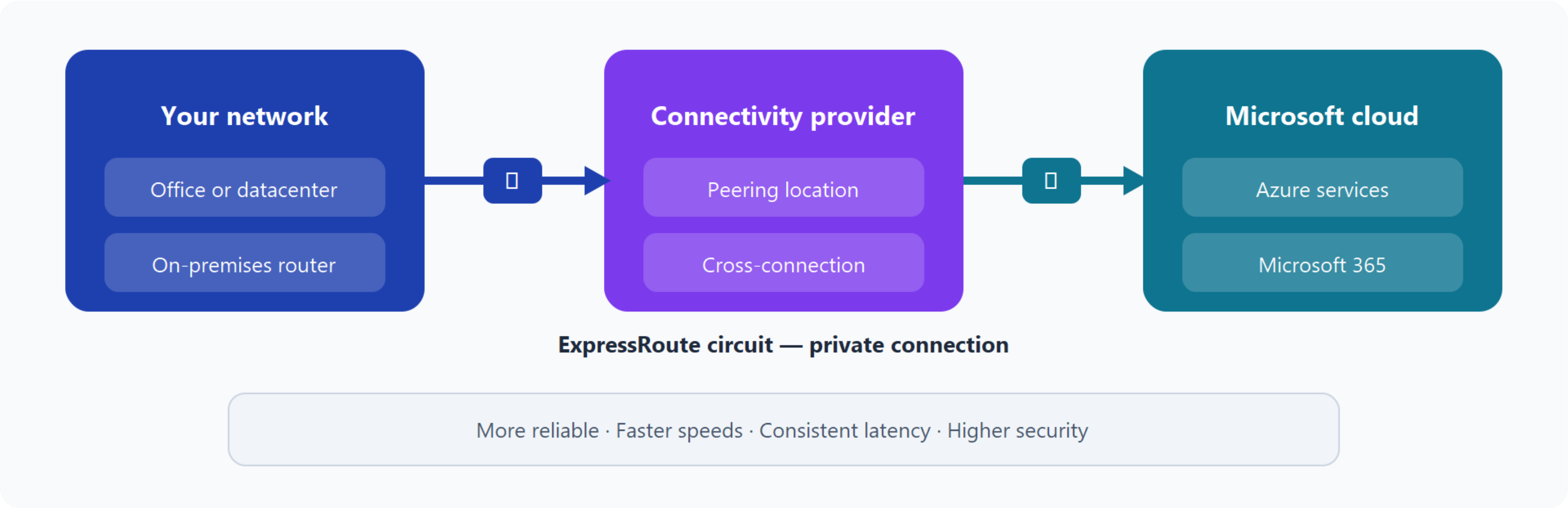
Azure virtual networks

- VNets provide isolation, segmentation, internet and cross-network connectivity
- Public endpoints have public IPs; private endpoints use IPs within the VNet address space
- Network Security Groups (NSGs) filter traffic with inbound/outbound rules by IP, port, and protocol
- VNet peering connects VNets (even cross-region) over the Microsoft private backbone
- On-premises connectivity: point-to-site VPN, site-to-site VPN, and Azure ExpressRoute

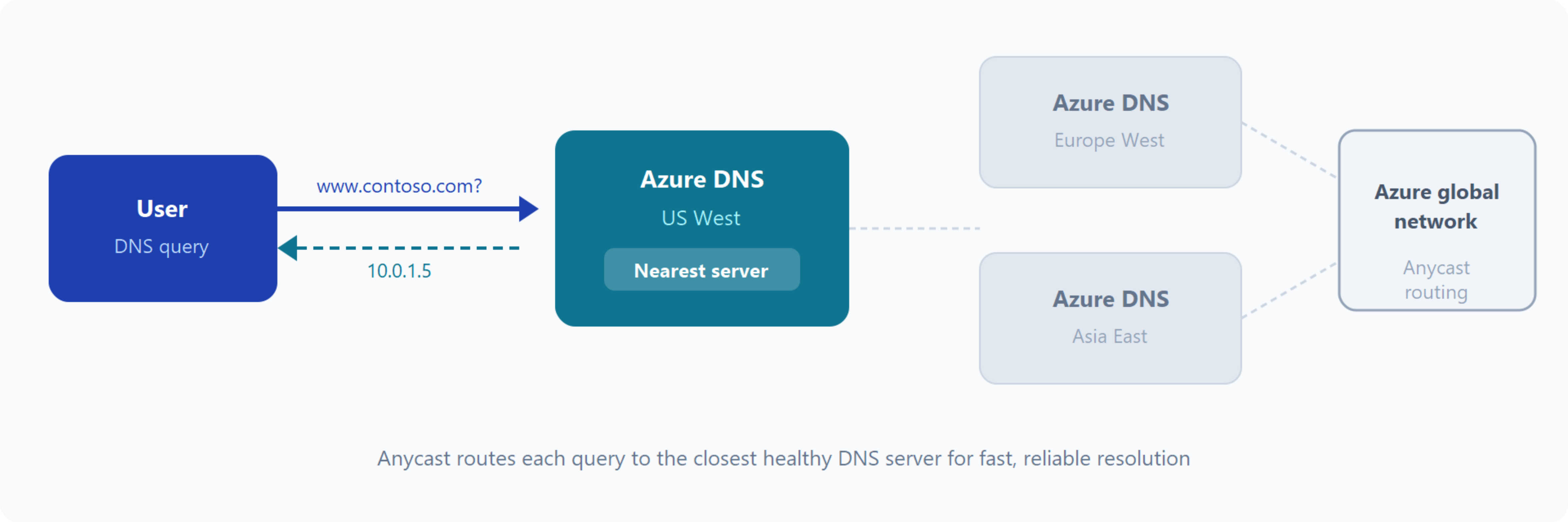
Azure VPN Gateway



Azure ExpressRoute



Azure DNS


User
DNS query
www.contoso.com?
10.0.1.5
Azure DNS
US West
Nearest server
Azure DNS
Europe West
Azure DNS
Asia East
Azure global network
Anycast routing
Anycast routes each query to the closest healthy DNS server for fast, reliable resolution

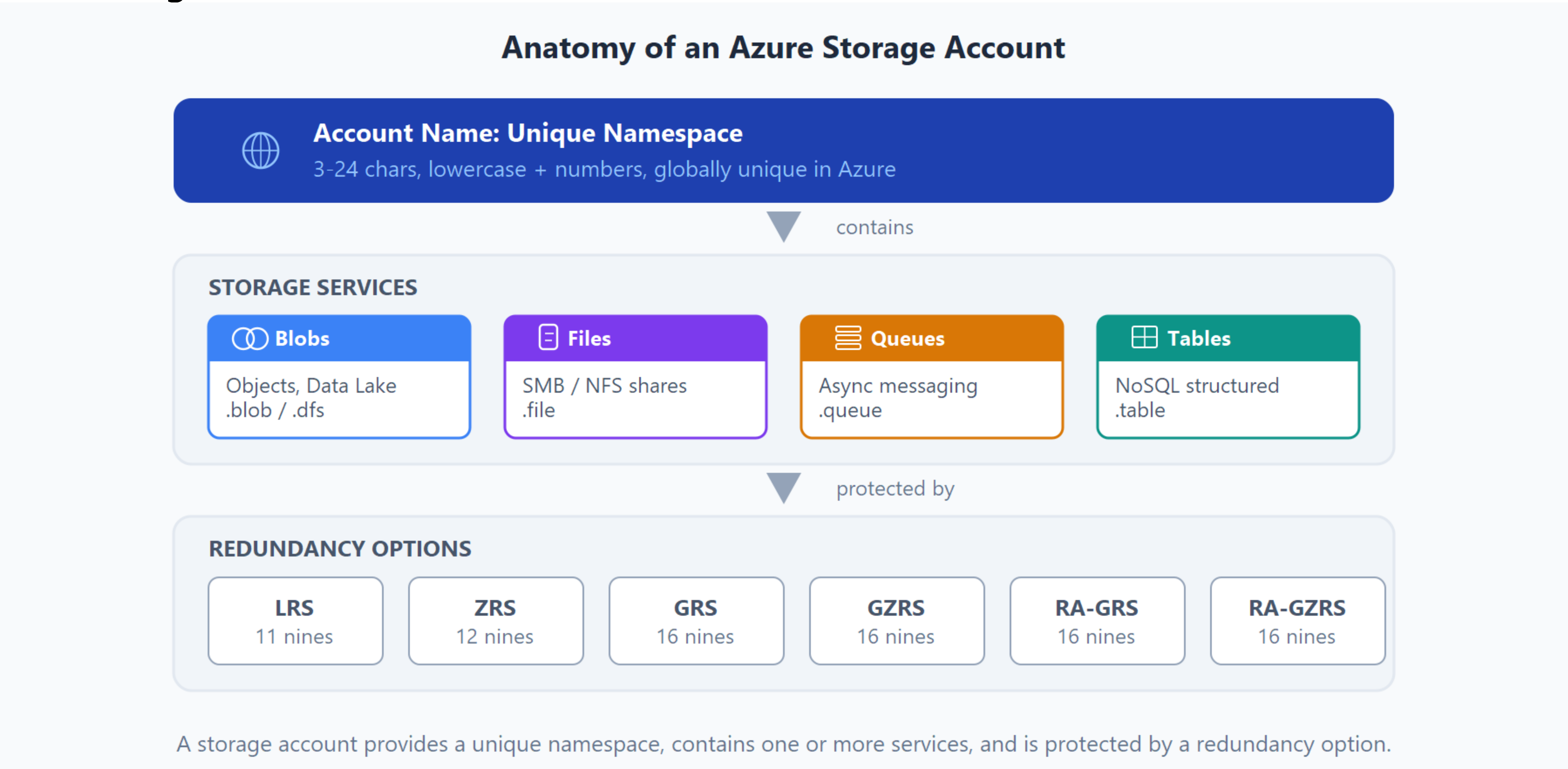
Knowledge check

1 **What is the primary purpose of a virtual network (VNet) in Azure?**

- ☐ Store unstructured data
- ☐ Provide isolated network communication for Azure resources
- ☐ Manage identities and access

Describe Azure storage services

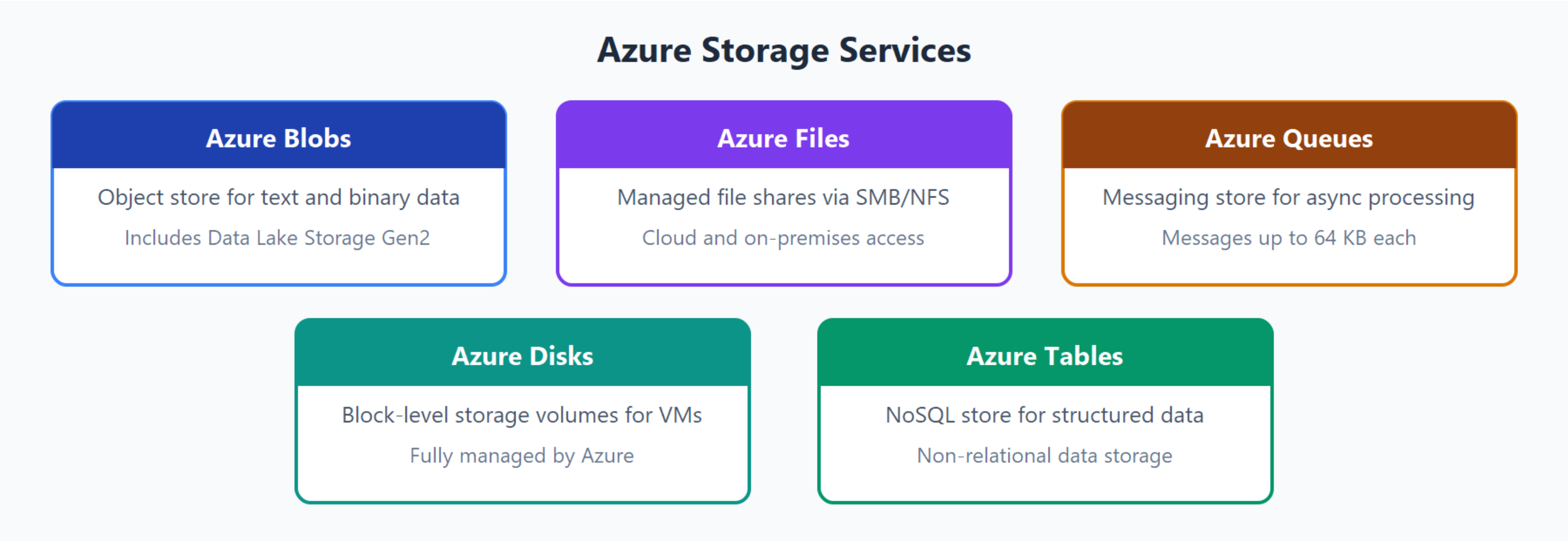
Azure storage accounts



Azure storage redundancy

- LRS replicates data 3x in a single datacenter (11 nines durability); lowest cost
- ZRS replicates across 3 availability zones in the primary region (12 nines durability)
- GRS uses LRS in both primary and secondary regions (16 nines durability)
- RA-GRS and RA-GZRS enable read access to secondary region data before failover

Azure storage services



Azure data migration options

Azure Migrate	Azure Data Box
Online	Physical
Real-time migration hub	**Offline bulk data transfer**
Single portal to assess and migrate on-premises infrastructure to Azure	Ship a secure storage device with up to 80 TB capacity
Discovery and assessment	One-time bulk migration
Server migration	Periodic large uploads
Database migration	Data export from Azure
Web app migration	NIST 800-88r1 disk wipe
ISV tool integration	End-to-end tracking in portal

Azure file movement options

Azure File Movement Tools

AzCopy

Command Line

- Upload and download files
- Copy between accounts
- One-way synchronization
- Cross-cloud provider

Sync Direction
One-way (source to destination)

Scripted / automated tasks

Storage Explorer

GUI App

- Graphical interface
- Windows, macOS, Linux
- Uses AzCopy on backend
- Manage blobs and files

Interaction Model
Manual upload / download

Visual browsing and management

Azure File Sync

Agent Service

- Bi-directional sync
- Cloud tiering
- SMB, NFS, FTPS protocols
- Multisite caching

Sync Direction
Bi-directional (automatic)

Hybrid file server scenarios

Each tool serves different file movement needs -- from scripted transfers to always-on hybrid sync

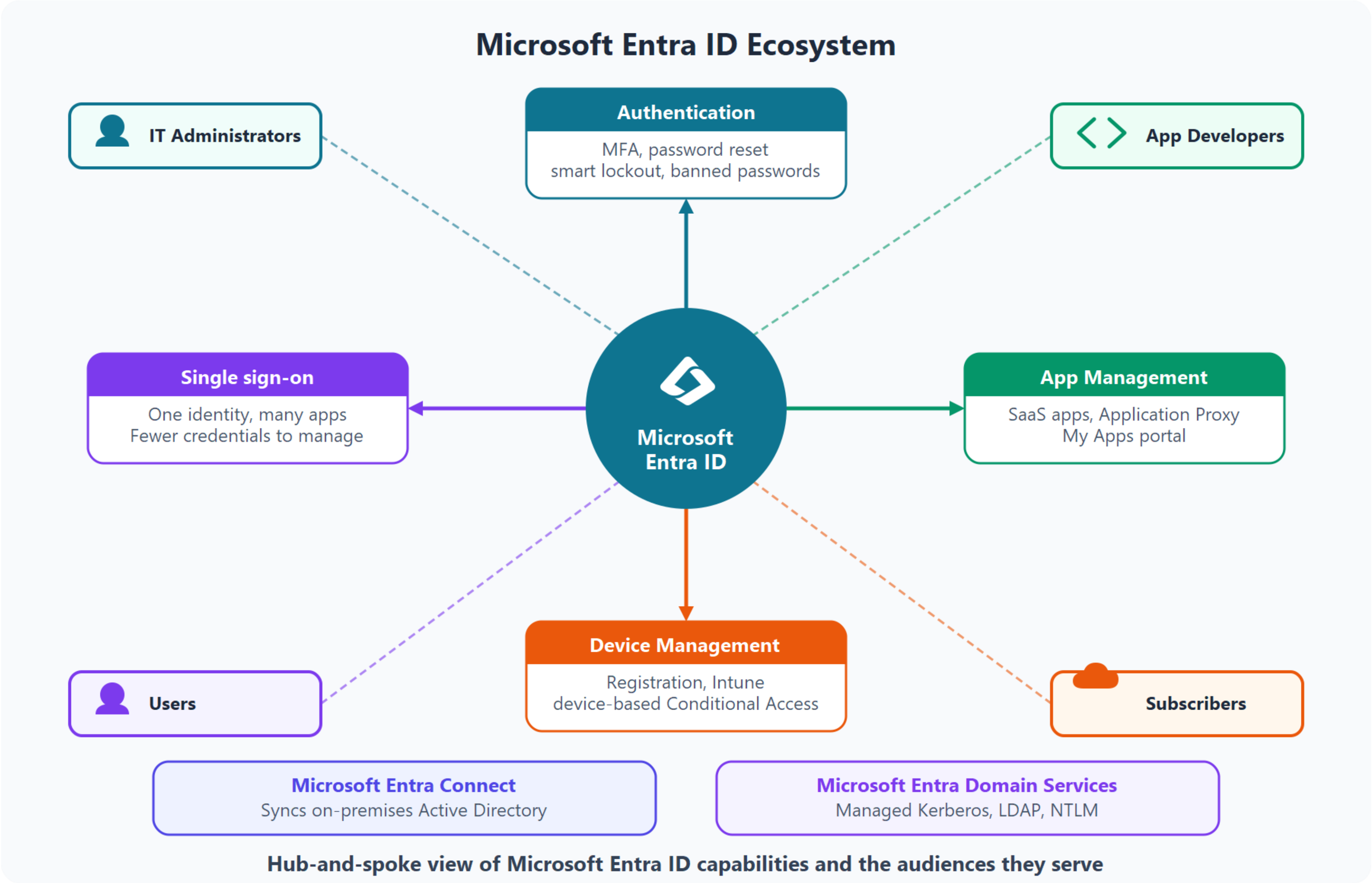
Knowledge check

1 **Which storage type is most appropriate for unstructured objects like images, backups, and logs?**

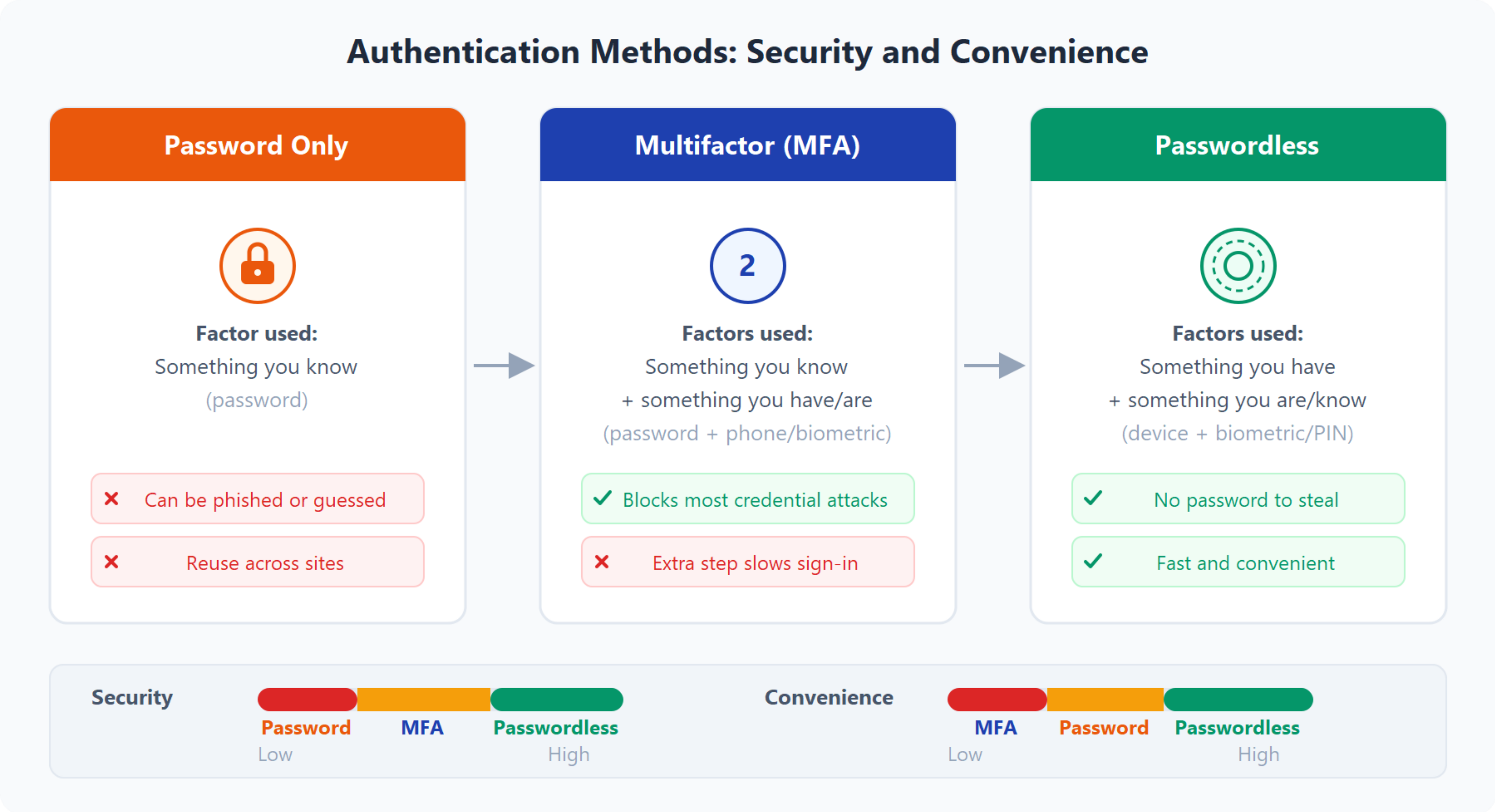
- ☐ Azure Queue Storage
- ☐ Azure Files
- ☐ Azure Blob Storage

Describe Azure identity, access, and security

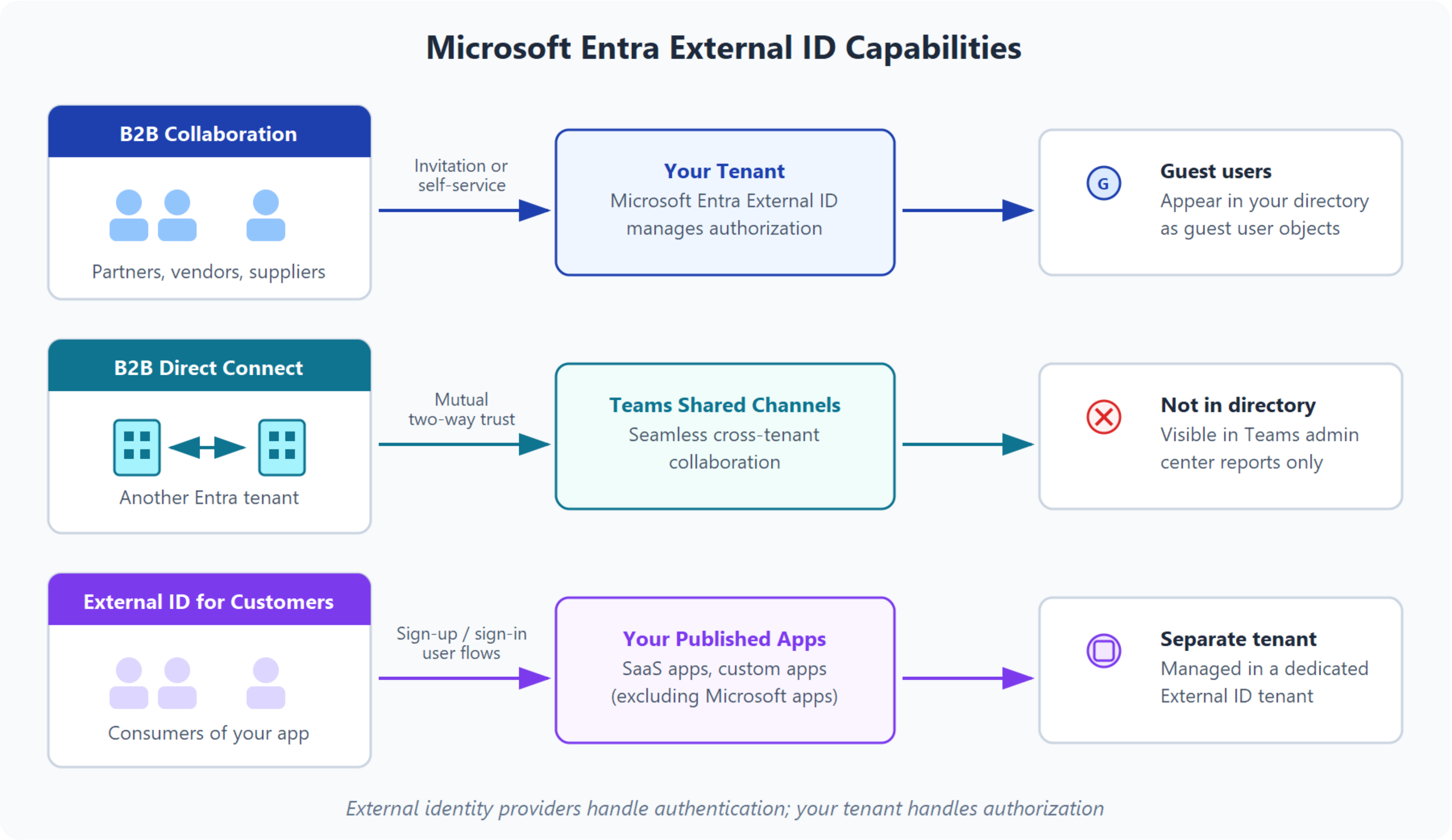
Microsoft Entra ID and Domain Services



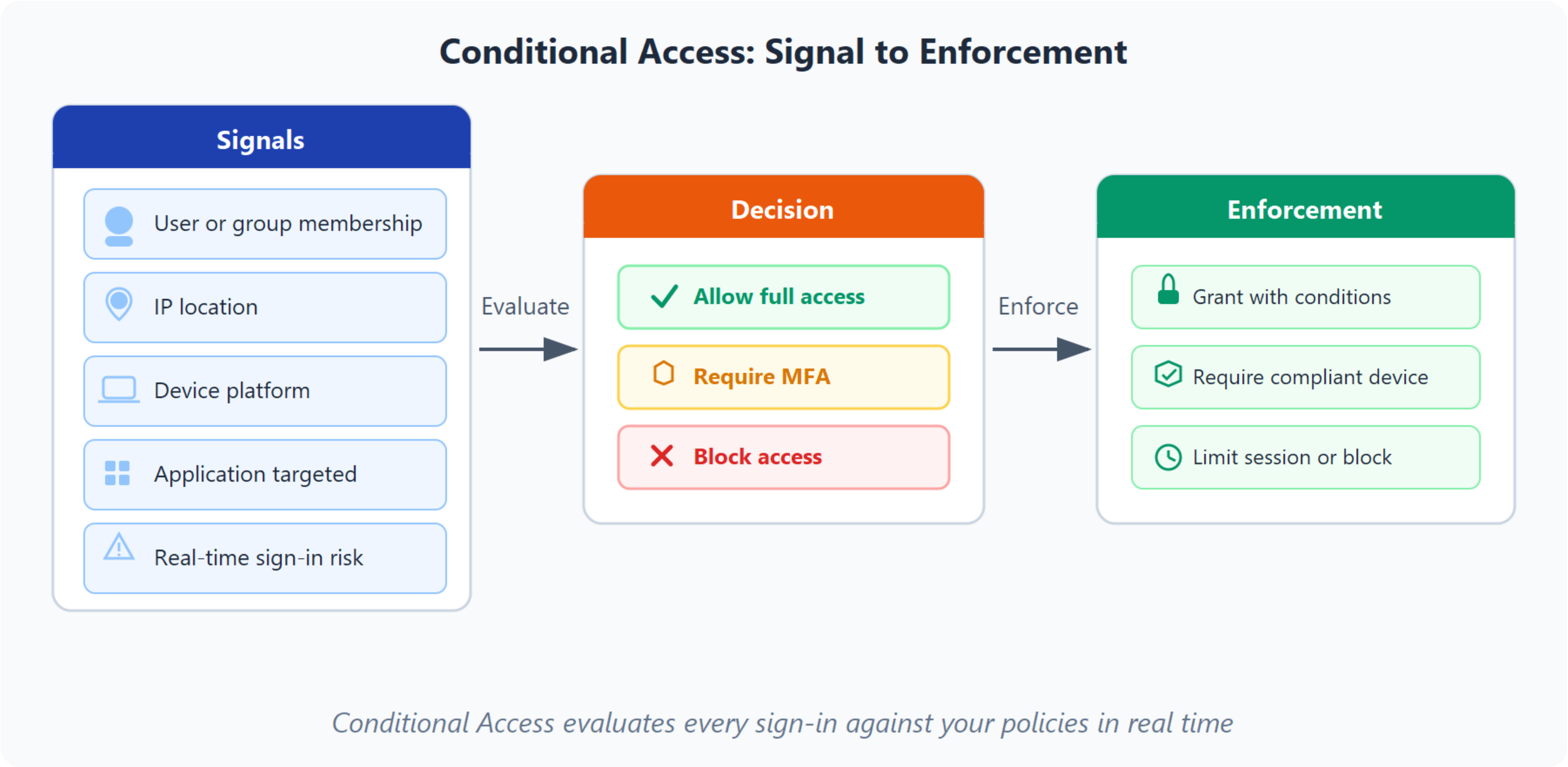
Authentication methods



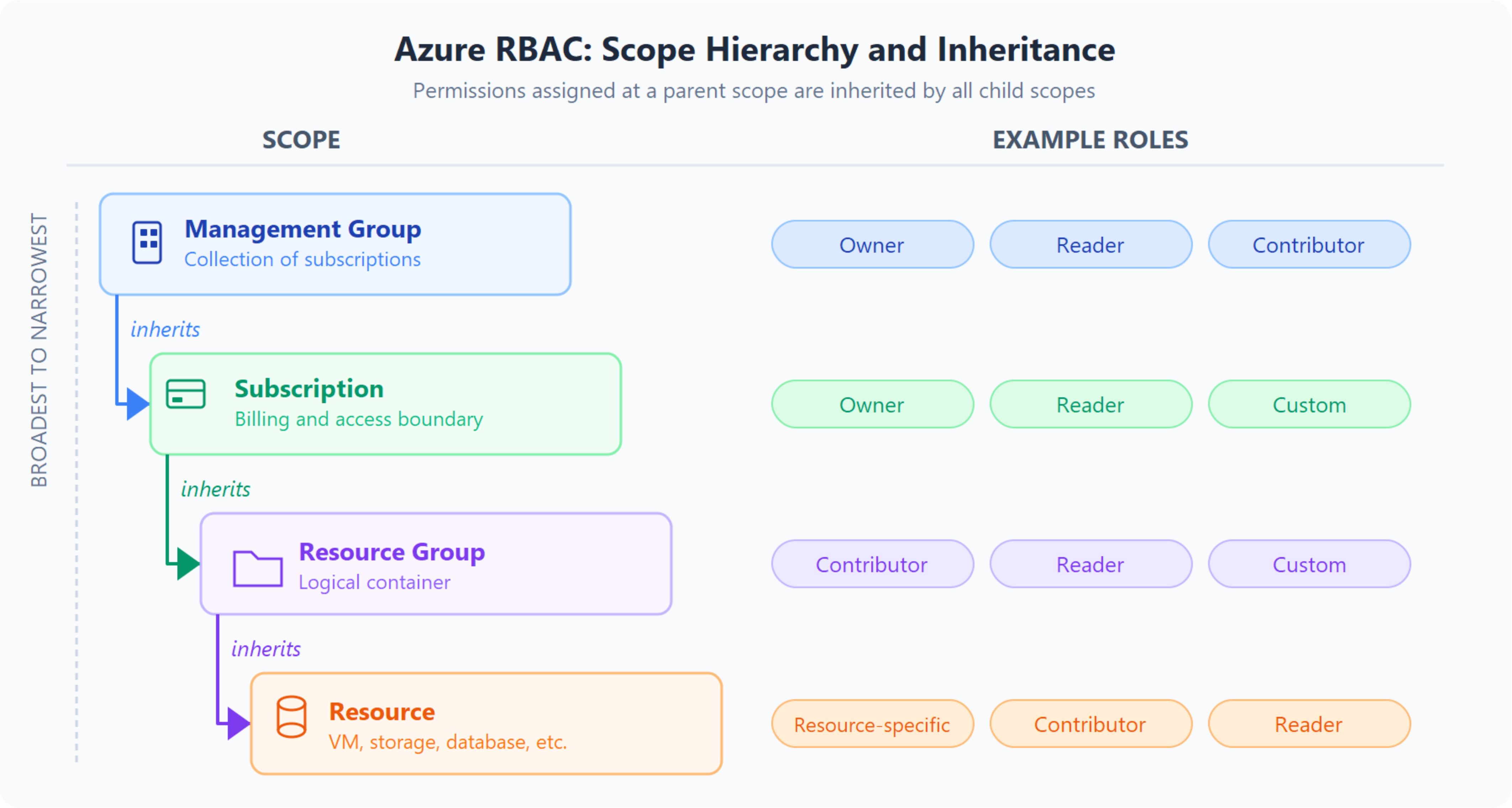
External identities



Conditional Access



Role-based access control (RBAC)



Zero Trust model

Zero Trust Guiding Principles

Verify explicitly

- Authenticate and authorize every request
- Evaluate all available data points
- Consider identity, location, device, service, and data classification

Least privilege access

- Limit access with Just-In-Time and Just-Enough-Access (JIT/JEA)
- Apply risk-based adaptive policies
- Protect data with policy-driven access controls

Assume breach

- Limit the potential impact and segment access
- Verify end-to-end encryption
- Use analytics for visibility, threat detection, and defense improvement

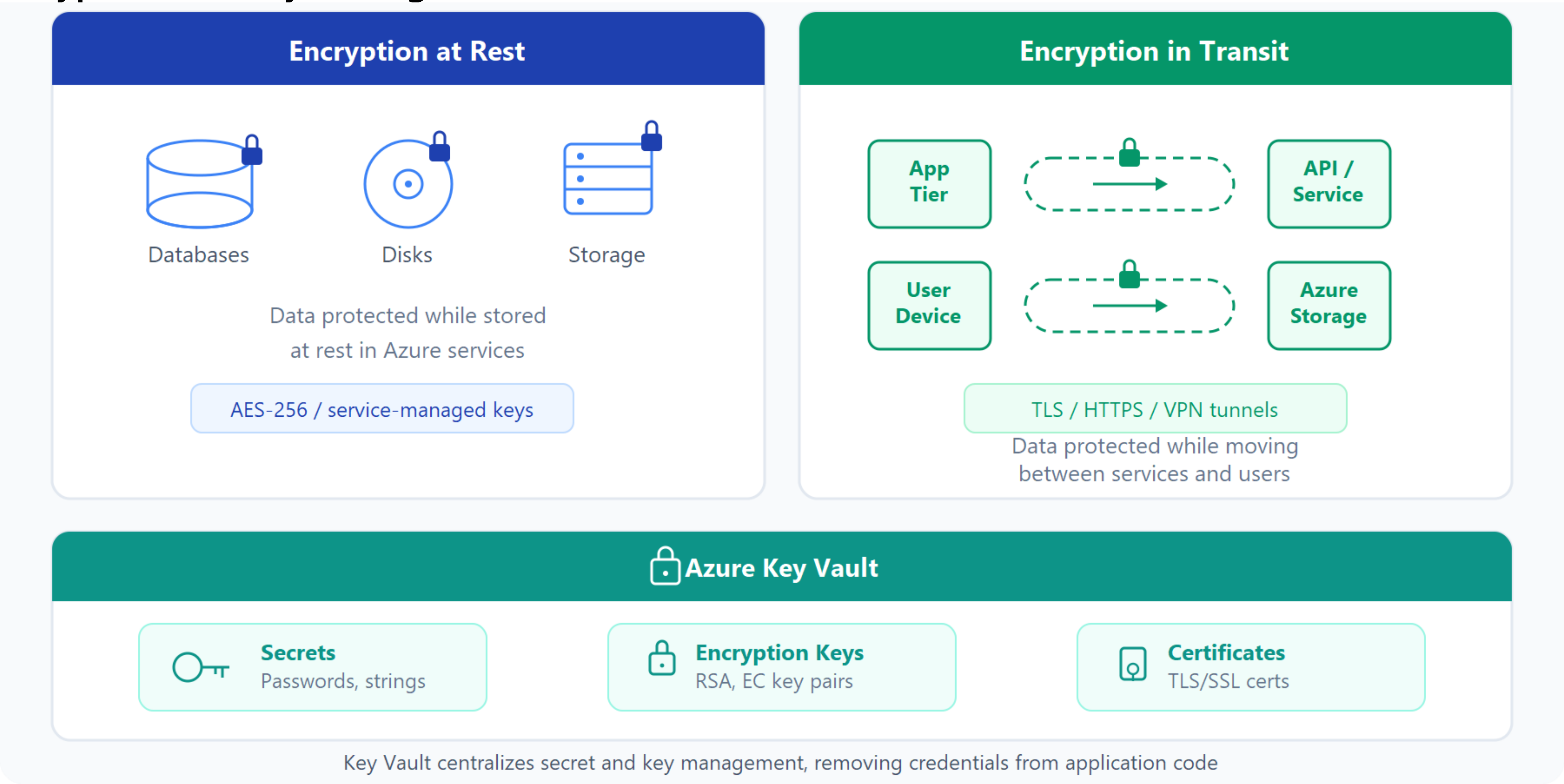
Defense in depth

Attacker progression

Layer	Role	Key Actions
Physical Security Outermost layer	Protect datacenter hardware	- Building access controls - Hardware security
Identity and Access Control who gets in	Secure identities, grant least access	- SSO and MFA - Audit events and changes
Perimeter Filter large-scale attacks	Protect against network-based attacks	- DDoS protection - Perimeter firewalls
Network Limit lateral movement	Segment and restrict communication	- Deny by default - Restrict inbound/outbound access
Compute Secure VMs and hosts	Patch systems, protect endpoints	- Secure VM access - Endpoint protection, keep patched
Application Secure by design	Eliminate vulnerabilities in code	- Secure development lifecycle - Store secrets in Key Vault
Data Ultimate target	Protect confidentiality, integrity, availability	- Encryption at rest and in transit - Access controls, regulatory compliance

No single layer provides complete protection -- multiple layers work together to slow and stop attacks

Encryption and key management



Microsoft Defender for Cloud

Three pillars of cloud security management

Continuously Assess

Know your security posture

Key Actions

- Identify and track vulnerabilities
- VM vulnerability scans
- Container registry assessments
- SQL server assessments
- Defender for Endpoint integration

Compute + Data + Infrastructure

Secure

Harden resources and services

Key Actions

- MCSB security policies
- Azure Policy integration
- Prioritized recommendations
- Secure score tracking
- Scope: mgmt group / sub / tenant

Policy-Driven Hardening

Defend

Detect and resolve threats

Key Actions

- Security alert generation
- Kill-chain attack correlation
- Remediation guidance
- Advanced threat protection
- Just-in-time VM access

Threat Detection + Response

Coverage: Azure-native | Hybrid (Azure Arc) | Multicloud (AWS, GCP)

Knowledge check

1 **Which Azure capability helps enforce conditions like MFA or location before sign-in is granted?**

☐ Role Based Conditional Access (RBAC)
☐ Conditional Access
☐ Azure Policy initiative

Summary

Learning Path 2: Describe Azure Architecture and Services

What you learned

- Regions and availability zones provide resiliency through geographic and datacenter-level fault isolation.
- Compute choice should match workload shape: VMs for control, AKS for orchestrated containers, Functions for event-driven execution.
- Storage, networking, and identity/security services form the base platform triangle for nearly every Azure solution.

End of presentation